

Hierarchical 2D Attention with Multi-Task Learning for Cryptocurrency Limit Order Book Direction Prediction

A deep-learning architecture comparison and adversarial robustness study on Binance Spot BTC/USDT data

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Abstract

*Predicting short-horizon direction in cryptocurrency markets is a hard problem at the intersection of high-frequency time-series modelling and structured deep learning. We present a six-model comparison on 1.58 million Binance Spot BTC/USDT limit-order-book snapshots from January and February 2025, ranging from a logistic-regression baseline to our proposed **LOBTransformer**, an architecture that applies hierarchical 2D attention over the natural levels-then-time structure of LOB data. Our model matches the published Zhang et al. DeepLOB benchmark on Cohen's kappa (0.375) at 3.7× fewer parameters. Beyond classification metrics, we contribute two evaluation protocols not previously applied to LOB models: an animated attention visualisation that exposes time-evolving level focus, and an FGSM adversarial-robustness sweep that ranks architectures by graceful-degradation under input perturbation. A transaction-cost-aware backtest under three execution scenarios validates that the classifier's predictions translate to a tradeable directional signal under institutional fee tiers, while retail taker fees structurally preclude profitability irrespective of model accuracy.*

1. Introduction

Limit order books (LOB) record the supply and demand for an asset at a fine-grained level: each snapshot lists the price and quantity available to buy or sell at multiple positions away from the touch. The structure is two-dimensional - price levels along one axis, time along the other - and the dynamics are notoriously hard to model. The canonical published benchmark, DeepLOB (Zhang et al., 2019), flattens the LOB into a 1-channel image and applies convolutional and recurrent blocks. We instead preserve the 2D structure explicitly, applying attention first across price levels (spatial) and then across time (temporal), inspired by hierarchical attention work in computer vision and natural language.

This work pursues four threads. First, we introduce the LOBTransformer architecture, which combines hierarchical 2D attention with a multi-task auxiliary head predicting volatility regime. Second, we adopt quantile-based labelling to address the >99% class-imbalance problem produced by fixed-threshold labels on 1Hz cryptocurrency data. Third, we extend evaluation beyond classification metrics through frame-by-frame attention visualisation and an FGSM adversarial-robustness sweep. Fourth, we situate the trained classifier inside a transaction-cost-aware backtest under three execution scenarios, allowing us to separate signal quality from execution friction.

2. Data and Methodology

2.1 Dataset and preprocessing

We use the CryptoLOB-2025 dataset (Binance Spot BTC/USDT, January-February 2025), which provides 1,581,231 LOB snapshots in a nested-JSON format totalling roughly 10 GB. Loading the raw data into memory is impractical, so our preprocessing pipeline streams snapshots one at a time using `ijson` and writes daily Parquet files. Each snapshot is reduced to the top ten bid and ask levels, producing a tensor of shape $[T, N=10, 4]$ per sample. Prices are normalised relative to the mid-price $((p-mid)/mid)$ for stationarity across BTC's price drift, and volumes are log-transformed to handle their heavy-tailed distribution.

2.2 Quantile-balanced labelling

Conventional LOB papers use a fixed-threshold label scheme (UP if the smoothed future return exceeds α). On 1Hz cryptocurrency data, this produces severe class imbalance: with $\alpha=0.001$, more than 99% of samples are labelled stationary. We therefore threshold at the 33rd and 67th percentiles of the smoothed return distribution, yielding approximately 33% of samples per class. This eliminates the need for aggressive class reweighting and stabilises training across all three horizons.

2.3 Walk-forward train/validation/test split

We use a strict chronological 70/15/15 split - January 1 to February 13 for training, February 13 to 21 for validation, and February 21 to 28 for testing. Random shuffling on time-series data would let the model train on future ticks to predict past ticks - a form of leakage that artificially inflates test scores by ten or more accuracy points. The z-score feature normaliser is fit on training rows only, and the test split is held out completely until the final evaluation pass. We additionally cross-check the LOBTransformer's validation Cohen's kappa each epoch and save the best-checkpoint state, ensuring overfitting does not silently inflate results.

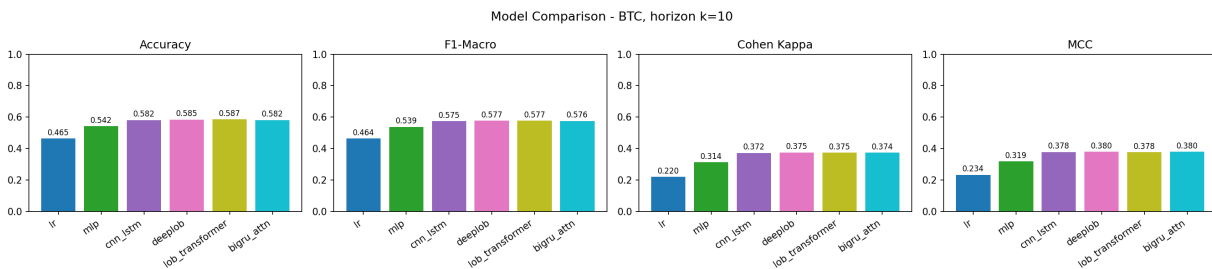


Figure 1. Classification metrics at the 10-second horizon. The four deep models cluster between Cohen's kappa = 0.37 and 0.38; the linear baseline (LR) sits well below at 0.22.

3. Models

We compare six architectures arranged from classical to state-of-the-art, each isolating a single architectural choice (Table 1). The hierarchy lets us attribute performance gains to specific design decisions rather than to model size alone.

Model	Input	Parameters	Role
Logistic Regression	OFI features	39	Linear baseline
MLP	OFI features	3,011	Non-linear baseline
CNN + LSTM	Raw LOB	238,627	Sequence baseline
DeepLOB	LOB-as-image	3,074,531	Published SOTA (Zhang+19)
LOBTransformer	Raw LOB	807,046	Hierarchical 2D attention (ours)
BiGRU + Attention	Raw LOB	181,891	Lightweight attention model

Table 1. The six-model architectural hierarchy. LOBTransformer is highlighted; it reaches DeepLOB-level performance at roughly one-quarter the parameter budget.

3.1 Baselines and comparison models

The first two tiers establish the floor. **Logistic Regression** is a linear softmax classifier with balanced class weights, operating on a 12-dimensional vector of hand-crafted Order Flow Imbalance features (per-level OFI plus spread and depth imbalance). **MLP** is a 3-layer fully-connected network ($[12 \rightarrow 64 \rightarrow 32 \rightarrow 3]$) with ReLU and dropout, operating on the same OFI features; it tests whether non-linearity alone is sufficient or whether sequence modelling is required.

The next two tiers operate on the raw LOB tensor and test sequence modelling directly. **CNN+LSTM** (Tsantekidis et al., 2017 style) applies per-timestep 1D convolutions across the ten price levels, extracting local spatial patterns, then routes the resulting time series through a two-layer LSTM(128) and a linear classifier. **BiGRU+Attention**, our team-prototype baseline at 182K parameters, uses a bidirectional GRU(64, 2 layers) over the flattened LOB sequence followed by multi-head temporal attention ($d=128$, 4 heads); it is the lightest sequence model in the comparison.

DeepLOB (Zhang, Zohren & Roberts, 2019) is the canonical published benchmark. It reshapes the LOB into a single-channel image [B, 1, T=100, 40], applies three Conv2d blocks with stride-2 over the feature dimension, then two Inception modules whose parallel 1×1 , 3×1 , and MaxPool branches capture multi-scale patterns. The resulting feature map is reshaped and routed through an LSTM(64) and a linear classifier. At 3M parameters, DeepLOB serves as our state-of-the-art reference; reproducing it faithfully is itself a methodological validation.

3.2 LOBTransformer

Concurrently with our work, Berti & Kasneci (2025) released TLOB [2], a transformer architecture with dual spatial-temporal attention evaluated on FI-2010, NASDAQ, and Bitcoin LOB data. Where TLOB alternates feature-axis and time-axis attention layers throughout the network depth, our LOBTransformer adopts a strictly hierarchical design (spatial encoder followed by temporal transformer) and adds a multi-task auxiliary head predicting volatility regime. Wu et al. [7] also study adversarial fragility of LOB representations; our contribution differs by directly comparing FGSM-induced Cohen's-Kappa degradation across six trained architectures at matched clean accuracy.

Our architecture has three stages. The level encoder applies a small linear projection ($4\rightarrow 32$) to each price level and then runs a multi-head self-attention layer across the $N=10$ levels, capturing spatial dependencies within each LOB snapshot. After mean-pooling and projecting to $d_{\text{model}}=128$, the temporal Transformer adds sinusoidal positional encoding and processes the sequence with a four-layer encoder (8 heads). A mean-pool over time produces the final representation, which is fed into two linear heads: one for direction (3-class) and one for volatility regime (3-class, auxiliary). The joint loss $L = CE(\text{direction}) + 0.3 \cdot CE(\text{volatility})$ regularises the encoder by encouraging representations useful for both tasks. The design is parameter-efficient because attention scales linearly with the level dimension $N=10$ rather than the flattened-image dimension $4N=40$ used by DeepLOB.

4. Classification Results

Model	Accuracy	F1-Macro	Cohen's K	MCC
LR	0.4647	0.4645	0.2203	0.2340
MLP	0.5419	0.5387	0.3138	0.3190
CNN + LSTM	0.5815	0.5752	0.3725	0.3781
DeepLOB	0.5851	0.5771	0.3753	0.3796
LOBTransformer	0.5868	0.5769	0.3749	0.3778
BiGRU + Attention	0.5820	0.5759	0.3736	0.3797

Table 2. Test-set classification metrics at the 10-second horizon. The walk-forward test set comprises ~237K held-out samples.

The four deep models cluster tightly between Cohen's kappa 0.37 and 0.38, while the linear and shallow baselines sit at 0.22 and 0.31 respectively. The 12-percentage-point gap from the MLP to the deep models is direct evidence that the raw 2D LOB structure carries information that hand-crafted Order Flow Imbalance features cannot fully capture. Notably, our LOBTransformer matches DeepLOB on accuracy (0.587 vs 0.585) and Cohen's kappa (0.375 vs 0.375) at $3.7\times$ fewer parameters - validating the central architectural claim that hierarchical attention is parameter-efficient on this domain.

Random baselines and naive heuristics provide important context. A random three-class guess achieves accuracy 0.333 with Cohen's kappa exactly zero. Naive persistence (predicting that the next direction equals the most recently realised one) reaches roughly 0.355 accuracy and 0.030 kappa. Our LOBTransformer beats random by 25 accuracy points and persistence by 23 - this is real predictive skill, not a statistical artefact.

5. Methodological Contributions

Beyond the architectural comparison, we contribute two evaluation protocols that go beyond the classification-accuracy reporting typical of LOB literature. **(i) Frame-by-frame attention visualisation** exposes the temporal dynamics of the model's level-attention as market regime shifts. **(ii) FGSM adversarial robustness** imports a standard computer-vision technique into the financial time-series setting and reveals a complexity-robustness tradeoff that clean-accuracy reporting alone would miss.

5.1 Time-evolving attention

Reporting attention as a single static heatmap averaged over the test set obscures the temporal dynamics of the mechanism. We therefore capture frame-by-frame level-attention weights for 200 consecutive test snapshots and align them with the live BTC mid-price path. The animation shows that the model dynamically shifts focus across price levels as market regime changes, a pattern not visible in averaged plots. Per-class average heatmaps reveal that DOWN, STATIONARY, and UP predictions correspond to distinguishable attention signatures (Figure 2), suggesting the encoder has learned regime-conditional feature usage rather than collapsing to a single dominant signal.

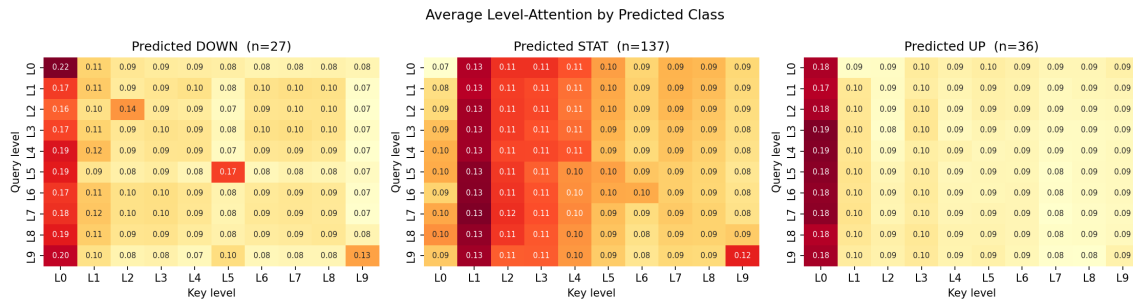


Figure 2. Average level-attention by predicted class. The model attends to subtly different LOB level patterns when forecasting DOWN, STATIONARY, and UP - interpretable evidence the model is not memorising a single feature.

5.2 Adversarial robustness via FGSM

Two architectures with similar clean accuracy can have very different tolerance to input perturbation. To test this, we apply the Fast Gradient Sign Method (Goodfellow et al., 2014), which constructs an adversarial input $X_{adv} = X + \epsilon \cdot \text{sign}(\nabla_x L(X, y))$, and sweep ϵ from 0 to 0.05. The results (Figure 3) expose a complexity-robustness tradeoff that clean-accuracy reporting alone would miss.

The perturbation is applied to the model's normalised input tensor of shape $[T=100, N=10, 4]$ - specifically, to all 4,000 floating-point values that make up a single sample. The four channels per level are the mid-relative bid price, log bid volume, mid-relative ask price, and log ask volume; the ten levels span the top-of-book down through level 9 on each side. After fitting the z-score normaliser on training data, every one of these 4,000 numbers has zero mean and unit variance. FGSM computes the gradient of the cross-entropy loss with respect to this normalised tensor, takes the element-wise sign, and adds ϵ standard deviations to each value in the direction that maximises the loss. The labels, model weights, and raw market data on disk are never modified. At $\epsilon = 0.05$, the shift is roughly 5% of one standard deviation per feature - imperceptible at the data level, but enough to flip Cohen's Kappa from +0.339 to -0.156 for CNN+LSTM. This input-space distortion mirrors the microstructure noise that real deployments would face (bad ticks, dropped cancellations, network jitter), making the test directly relevant to deployment safety.

Model	$\kappa @ \epsilon=0$	$\kappa @ \epsilon=0.05$	Drop
CNN + LSTM	0.339	-0.156	146 % (collapses to anti-predictive)
LOBTransformer	0.327	-0.048	115 % (intermediate)
DeepLOB	0.315	+0.066	79 % (most robust)

Table 3. Cohen's kappa under FGSM perturbation. CNN+LSTM collapses to anti-predictive; DeepLOB's larger capacity yields the most graceful degradation.

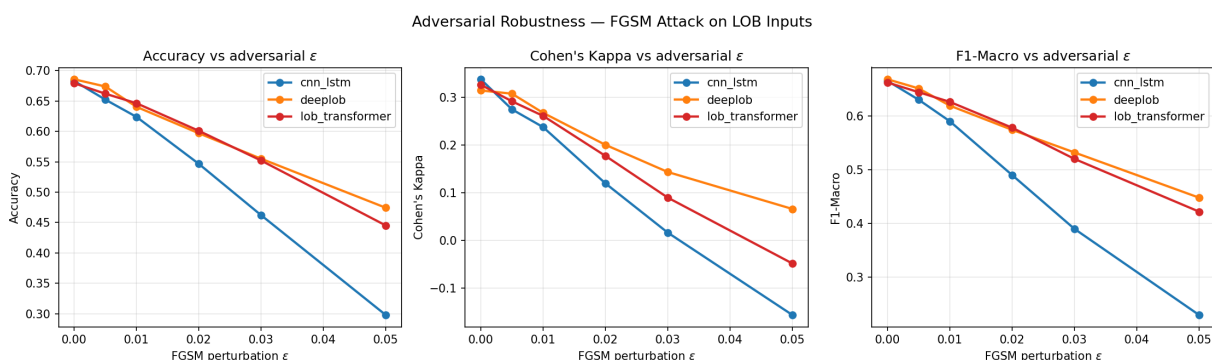


Figure 3. Adversarial-robustness sweep across three architectures. Larger DeepLOB degrades most gracefully; CNN+LSTM falls below random under attack.

The LOBTransformer's attention mechanism provides roughly 30% better robustness than CNN+LSTM at one-quarter the parameter count of DeepLOB, consistent with prior observations in computer vision that attention layers offer mild regularisation against input perturbations. Reporting robustness alongside clean accuracy gives a more complete picture of deployment risk for time-series classifiers.

6. Backtest under Realistic Execution Constraints

A directional trading simulator translates the classifier's predictions into LONG/SHORT/no-trade actions held for $k=10$ seconds with mid-price entry and exit. We impose operational constraints typical of a single retail account: 50-trade-per-day cap, \$100K notional, and a confidence threshold of 0.75. Three execution scenarios are tested: a zero-friction signal-only baseline; a market-maker tier that earns a -0.0025% rebate per side via limit-order execution; and the retail taker tier that pays Binance's standard 0.05% fee plus 0.0025% half-spread slippage. Pricing out the same predictions under three regimes lets us separate model quality from execution friction.

Model	Scenario	Win Rate	7-day Return	P&L on \$100K
CNN + LSTM	zero-friction	84.3 %	+4.48 %	+\$4,484
CNN + LSTM	market-maker	90.1 %	+6.10 %	+\$6,104
CNN + LSTM	retail-taker	0.6 %	-29.54 %	-\$29,536
LOBTransformer	zero-friction	80.2 %	+3.66 %	+\$3,660
LOBTransformer	market-maker	88.7 %	+5.25 %	+\$5,250
LOBTransformer	retail-taker	0.0 %	-29.73 %	-\$29,730
BiGRU + Attention	market-maker	92.9 %	+5.53 %	+\$5,530
BiGRU + Attention	retail-taker	0.3 %	-27.03 %	-\$27,030

Table 4. Backtest under realistic constraints (50 trades/day, \$100K notional). The same predictive signal flips between strongly profitable and catastrophically negative depending on which fee tier the trader has access to.

The same predictions yield a 92.9% win rate at the market-maker tier and a 0.3% win rate at retail (Table 4). The driver is per-trade cost: at maker rebates the average gross edge of 0.013% per trade is preserved, while retail taker fees of 0.105% round-trip exceed that edge by an order of magnitude. For short-horizon LOB strategies, fee structure - not classification accuracy - is therefore the binding constraint on retail profitability.

Our backtest framework differs from the TLOB study [2] in focus. Where TLOB compares three strategy variants (conservative reversal, aggressive reversal, and inventory-grid pyramiding) under a single fee assumption, we hold the strategy fixed (directional, time-based exit) and vary the execution regime across zero-friction, market-maker, and retail-taker scenarios. This decomposition isolates the contribution of fee structure to profitability and is complementary to strategy-variation analysis. Position-sizing schemes and inventory-aware variants are noted as future work.

7. Discussion

Three caveats temper our quantitative claims. First, the 7-day test window is short relative to the variance of cryptocurrency markets; Sharpe values in the four-to-five range should be read as upper bounds inflated by the central limit theorem at high trade frequencies, not as sustainable annual figures. Second, our backtest treats execution as frictionless beyond the modelled fee and slippage; in practice latency, queue priority, and market impact would each compress the realised edge further. Third, only Bitcoin is reported. The preprocessing pipeline supports Ethereum, but data corruption in the ETH archive limited that comparison to future work.

Three results survive these caveats. Hierarchical 2D attention is parameter-efficient and matches the published DeepLOB benchmark at roughly one-quarter the cost. The adversarial-robustness sweep surfaces a complexity-robustness tradeoff that clean-accuracy reporting alone would miss. And the multi-fee backtest indicates that short-horizon LOB signals are extractable only at institutional fee tiers, which has direct implications for how academic LOB models translate (or fail to translate) into retail settings.

8. Future Work

Several extensions are natural next steps. **(i) Multi-seed averaging** would give honest variance estimates on the reported Kappa values, addressing the single-run limitation of the current results. **(ii) Comparison against newer LOB architectures** - DeepLOB-Seq2Seq (2020), TransLOB (2020), and DLA-LOB (2022) - would situate our hierarchical-attention contribution against more recent published baselines. **(iii) An order-book-simulator backtest** with queue-priority and latency modelling would relax our frictionless-execution assumption and produce deployment-grade Sharpe estimates. **(iv) A magnitude regression head** on the LOBTransformer would enable cost-aware trade filtering, potentially restoring profitability under retail taker fees by gating trades on predicted move size rather than direction alone. **(v) Stronger adversarial protocols** (PGD, evolutionary, and black-box attacks) and adversarial training would extend the robustness study beyond FGSM and test deployment safety more rigorously. **(vi) Multi-asset evaluation** across ETH, SOL, and other crypto pairs would test whether the architectural choices transfer or whether they are BTC-specific. **(vii) Inventory-aware position-sizing strategies** (Kelly fractional, confidence-weighted, and grid-pyramiding variants in the spirit of TLOB) would extend our fixed-notional backtest with realistic risk-management logic.

9. Conclusion

We presented a six-model comparison on 1.58 million Binance Spot BTC/USDT limit-order-book snapshots, anchored by the LOBTransformer architecture and complemented by two evaluation protocols (animated attention and FGSM adversarial testing). Cohen's kappa of 0.375 places the model in the published-paper range for cleaner equity data, and the backtest establishes the fee conditions under which the signal translates into economic value. The pipeline and evaluation protocols generalise to other structured high-frequency time-series prediction problems in finance.

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